

Generative AI Driven Real-time Open-set Object Detection for Mobile AR Applications

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Abstract—Mobile augmented reality (MAR) faces significant challenges in resource-constrained environments, where on-device object detection models are limited in size and class coverage, struggling to detect out-of-range objects in open-set environments. This degrades user quality of experience while existing approaches relying on manual data collection and annotation are inefficient for real-time demands. We propose a generative AI-driven framework for real-time open-set object detection in MAR applications. By leveraging stable diffusion-based generative models, our method synthesizes annotated images to enable efficient real-time incremental learning. The framework's two-stage architecture includes a frozen open-set region proposal network to identify regions of interest and a dynamic hot classification module to classify new object classes, enabling rapid updates and extended class coverage.

Experiments demonstrate a 10x improvement in incremental learning speed, reducing process time to under two minutes. This innovation improves real-time, accurate object detection, significantly enhancing MAR applications and user QoE within mobile device constraints.

Index Terms—MAR, Object detection, Incremental learning, Generative AI

I. INTRODUCTION

Mobile Augmented Reality (MAR) applications have gained significant attention in recent years due to their ability to seamlessly integrate virtual objects into physical environments, enabling interactive and immersive user experiences. With advancements in on-device neural network models and MAR development platforms [1], MAR applications have become increasingly accessible on mobile devices across various scenarios, including tourism, advertising, and entertainment [2]. Among the core functionalities of MAR applications, object detection is one of the most widely utilized features, facilitating the recognition and localization of objects within the user's view [3]. To support real-time object detection, existing MAR frameworks deploy object detection models directly on mobile devices, ensuring efficient and responsive performance [4].

One of the challenges faced by on-device object detection models is the inability to recognize and detect unknown objects in dynamic environments. For object detection, one of the most widely used datasets, the Microsoft COCO dataset, contains 80 object classes [5], while LVIS contains up to 1,200 classes [6]. On one hand, in open-set environments where many MAR applications operate, objects in dynamic and unpredictable scenes are likely to exceed the predefined limit

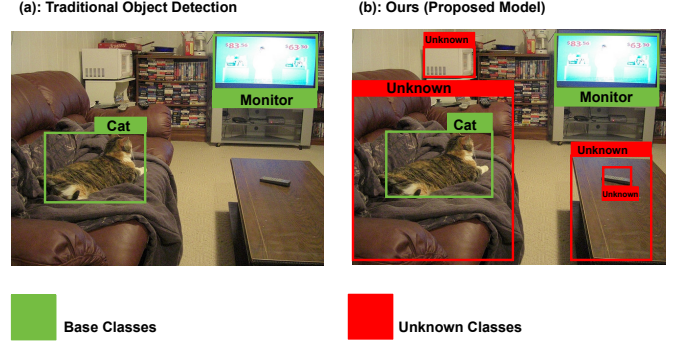


Fig. 1. The demonstration of open-set object detection.

of 80 classes [7]. Consequently, on-device object detection models trained on smaller datasets often fail to recognize objects from unknown or novel classes. As shown in Fig. 1(a), existing frameworks are limited to detecting only the base classes. On the other hand, models trained on these larger datasets are unsuitable for deployment on mobile devices due to their high computational complexity. Even when leveraging these more powerful models by offloading the tasks to edge servers, open-set environments inevitably present objects from novel classes, further highlighting the limitations of current approaches. In this work, we aim to enable on-device object detection models to recognize and detect unknown objects in dynamic environments, as depicted in Fig. 1(b).

With the advancement of large model techniques, state-of-the-art object detection models, such as DINO-X [8], can partially address the open-set object detection (OSOD) problem. However, when an unknown object appears in a scene, DINO-X requires users to input prompts, such as the object's class name or location, to identify and localize the target object. This manual intervention disrupts the natural flow of user interaction in MAR applications. For example, users cannot provide the name of an unknown object to directly recognize it. This limitation also affects edge computing-based object detection frameworks. On-device object detection models lack the capability to recognize failures in detecting unknown objects, and frequent task offloading results in reduced user Quality-of-Experience (QoE) [9].

In addition to requiring user involvement, two other approaches aimed at addressing the OSOD problem are **Few-Shot Object Detection (FSOD)** and **Object Detection In-**

Incremental Learning [10], [11]. FSOD aims to learn new object classes from a limited number of annotated instances. Initially, base classes are trained, followed by fine-tuning using annotated data from novel classes. However, the fine-tuning process includes image data from all base classes, which significantly increases training time and makes this approach unsuitable for real-time OSOD scenarios. Object Detection Incremental Learning integrates novel classes into an existing object detection model using only data from the novel classes. However, the training efficiency of current incremental learning frameworks is heavily dependent on the number of classes and the data quality of the novel classes. For example, adding one novel class to a model with 20 base classes may take only two minutes, but extending the model to include five novel classes with 60 base classes can take several hours.

In this paper, we address the aforementioned challenges and present the first end-to-end real-time open-set object detection incremental learning framework. Our main contributions are as follows:

- We conduct a comprehensive analysis of the characteristics and challenges associated with open-set object detection in MAR applications, providing valuable insights into this emerging research area.
- We propose a novel two-stage object detection framework designed for efficient model incremental learning. This framework incorporates a frozen open-set Region Proposal Network (RPN) and a dynamic object classification module. It is much faster than existing FSOD and object detection incremental learning approaches.
- To tackle the issue of poor-quality datasets, we leverage generative AI models to synthesize high-quality images and annotations for novel classes, ensuring consistent data quality.
- We introduce a seconds-level end-to-end object detection incremental learning framework. Extensive evaluations demonstrate that our approach enables incremental object detection at unprecedented speeds, achieving real-time performance.

II. RELATED WORK

In this section, existing work related to deploying OSOD at the mobile edge is introduced.

A. Few-Shot Object Detection and Incremental Learning

FSOD aims to learn new object classes from a limited number of annotated instances. Unlike traditional object detection methods, which rely on the costly and time-intensive process of collecting and annotating large volumes of data to train accurate detectors [12]. FSOD leverages existing class knowledge to facilitate the learning of new classes with only a few examples [13], [14].

Despite its advantages, a key limitation of FSOD for detecting novel classes is its dependency on fine-tuning on the entire annotated dataset, which includes both base and novel classes [15]. Although FSOD requires only a few instances for

novel class training, the process of fine-tuning the full dataset can take several hours. This lengthy training process makes FSOD unsuitable for fast model updates in dynamic, real-time environments.

Incremental learning has been widely applied in image recognition tasks, enabling the integration of novel classes into existing recognition models using only data from the new classes. However, its application to object detection remains relatively underexplored. RILOD [10] presents a lightweight incremental learning model designed to seamlessly incorporate novel object classes into existing object detection frameworks without requiring access to the original training data. By leveraging knowledge distillation and feature representation alignment, RILOD mitigates catastrophic forgetting and enhances recognition of new classes, ensuring robust performance across both previously learned and newly introduced classes. Sylph [11] introduces a hypernetwork-based framework for incremental few-shot detection (iFSD). This framework employs a class-agnostic base detector for object localization and a hypernetwork to dynamically generate class-specific binary classifiers for novel classes.

Despite these advancements, incremental learning frameworks face significant challenges related to training efficiency and model accuracy. As the number of novel classes increases, the training time within a unified framework grows, and the overall accuracy of the object detection model tends to decline. Our experimental results indicate that the accuracy of novel classes degrades significantly when their number exceeds fifteen.

B. Data Collection and Annotation

For accurate object detection, digital image datasets play a critical role, as they enable the model to learn object features, process them, and localize objects within the frame while classifying object types. While many open-source datasets for object detection are available, synthetic datasets can be leveraged to introduce new object classes and enable adaptation to open-set environments. Generative AI models have significantly advanced the field of image synthesis, particularly in text-to-image generation. Diffusion models, such as DDPM and Stable Diffusion [16], [17], have demonstrated remarkable capabilities in producing high-quality, high-resolution images. These models employ a stepwise denoising process, gradually refining random noise into coherent images, offering a compelling alternative to traditional generative approaches like GANs and VAEs [18], [19]. Despite these breakthroughs, diffusion models face notable challenges related to their substantial computational and memory requirements, especially for large-scale and high-resolution image generation tasks. To address these limitations, various optimization techniques, including pruning, knowledge distillation, and model compression, have been proposed to reduce model size and computational overhead, facilitating deployment on resource-constrained devices.

To the best of our knowledge, no existing work has utilized generative AI models to synthesize datasets specifically for OSOD incremental learning.

C. Edge Computing Detection

Edge computing, which utilizes hierarchical computing resources distributed across mobile edge devices, edge servers, and remote clouds, has emerged as a transformative solution to address the computational challenges of object detection tasks in resource-constrained environments. For example, powerful object detection models like DINO-X, while delivering high performance, are computationally intensive and memory-demanding, making them impractical for deployment on resource-limited edge devices such as AR glasses, which often lack the processing power and memory required to efficiently run large-scale models. To overcome this limitation, edge computing detection frameworks combine lightweight on-device object detection models with more computationally intensive models running on powerful edge servers that have greater processing capabilities [9]. In this approach, object detection tasks are initially performed on the mobile device. If the system’s offloading decision algorithm determines that offloading the task to a more powerful edge server would significantly improve the user QoE, the captured frame is sent to the edge server. The server then performs the object detection task and sends the result back to the mobile device. This approach leverages the low latency and responsiveness of task execution on the mobile device while taking advantage of the edge server’s ability to run more accurate and complex models. The key advantage of this edge-based system is that it combines the best of both worlds: fast, low-latency processing on the mobile device and high-accuracy detection from the server. However, task offloading has its drawbacks. The process of offloading frames to the server and feeding the detection results back to the device introduces additional network latency, which can diminish the overall user experience, especially in real-time applications where responsiveness is critical.

To mitigate the impact of task offloading, our work focuses on enhancing the capabilities of on-device models. By improving on-device models, we aim to reduce the need for offloading, thus minimizing network latency and improving overall system performance in edge-based MAR applications.

III. SYSTEM FRAMEWORK

In this work, we propose the first end-to-end real-time open-set object detection incremental learning framework. As shown in Fig. 2, the proposed framework consists of three main modules: the on-device object detection module in the gray dotted box, the generative AI module for data collection and annotation in the red dotted box, and the class incremental learning module in the green dotted box. The on-device object detection module operates on the mobile device, while the other two modules are deployed on the edge server. A lightweight version of these two modules can also be

TABLE I
CONFIGURATION OF GENERATIVE MODELS

	BK-SDM	SD 3.5	Layout-diffusion
Speed (s/image)	0.82	9.7	3.2
#Param	0.66B	8.1 B	569 M
Model Size	654.9M	788.48 M	2.1 G

TABLE II
DE-ViT WITH REAL AND GENERATED IMAGES FOR 4 SESSIONS

Image Type	Shot	AP_{base}	AP_{novel}	AP_{all}
Real Images	10 shot	29.4	34.0	37.0
	30 shot	29.5	34.0	37.2
Generated Images	10 shot	26.21	33.62	33.62
	30 shot	27.0	34.41 ↑(+0.41)	34.41

deployed on the mobile device, ensuring flexibility in resource-constrained environments.

Object Detection Module. We design a Faster R-CNN [20]-based object detection framework that includes an image feature extractor, a frozen open-set Region Proposal Network (RPN), and a hot classification module. Unlike the standard Faster R-CNN, the introduced novel open-set RPN is capable of proposing regions of interest (RoIs) for thousands of different object classes. The open-set RPN is frozen during the model’s incremental learning process and is not updated, which helps overcome the limitations of conventional object detection incremental learning approaches. This design significantly accelerates the incremental learning speed and improves overall efficiency. The open-set RPN detects all potential RoIs in the input image, which are then passed to the classifier for classifying the objects within the bounding boxes.

Generative-AI Module. The output from the object detection module includes objects from both base classes and unknown objects from novel classes. These unknown objects are forwarded to a vision large model (Region CLIP [21]), which recognizes the classes of these previously unseen objects. A generative model is then employed to generate high-quality images and annotations for these novel classes. Among various generative models, such as layout-image generation and stable diffusion [22], [23], we adopt stable diffusion models in our framework due to their compatibility with our two-stage object detection incremental learning process.

Class Incremental Learning Module. To achieve fast and efficient object detection incremental learning, we incorporate the state-of-the-art class incremental learning module, TEEN [24]. This module facilitates the hot updating of the classifier in the object detection module. TEEN generates a new classifier using images and annotations of novel classes along with the prototype of the base classifier. The entire model incremental learning process takes only a few seconds in a standard setup, ensuring minimal disruption and fast adaptation to new classes.

In terms of data generation quality and speed and the efficiency of object detection incremental learning, the proposed generative AI-driven two-stage framework, to the best of our knowledge, achieves optimal performance.

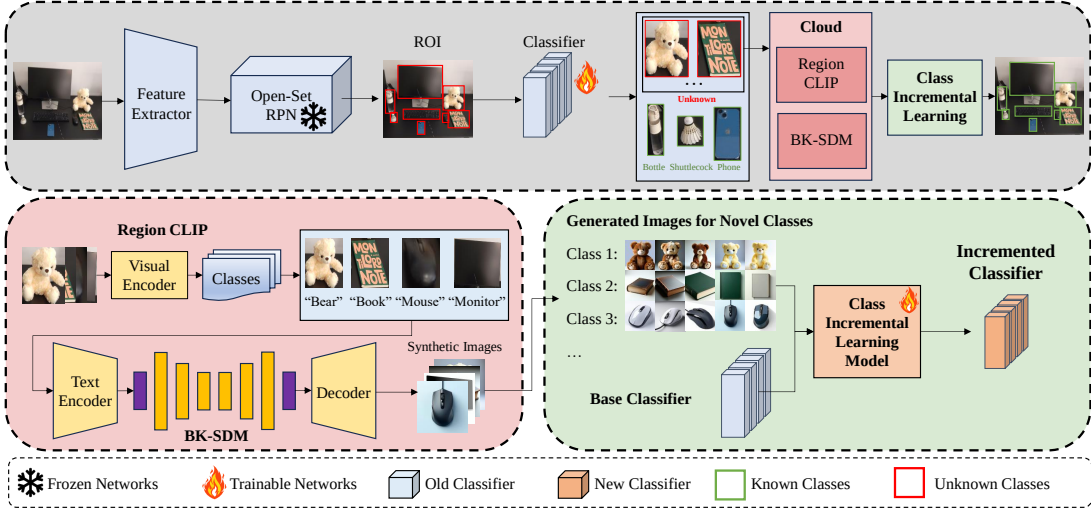


Fig. 2. The proposed end-to-end real-time open-set object detection framework.

IV. PERFORMANCE EVALUATION

In this section, we provide a detailed analysis of the proposed framework and compare its performance with existing approaches.

A. Experiment Settings

Generative-AI model. To generate images of novel classes, we evaluate three image generation models: the Layout-Diffusion model, Stable Diffusion (SD) 3.5-large, and the lightweight BK-SDM-small [22], [23]. While both SD and BK-SDM generate a single object per image, the Layout-Diffusion model has the capability to generate images containing multiple objects from different classes. This feature can enhance training by providing a richer set of diverse objects, which may improve the accuracy of the object detection model. The Layout-Diffusion model generated results contain both bounding boxes and images that are essential for FSOD models but cannot be directly integrated with the proposed two-stage incremental learning approach.

Object detection model. We use Faster R-CNN as the backbone for the object detection model. The MSCOCO dataset is employed for training, with 60 classes selected as base classes and an additional 20 classes designated as novel classes. The frozen open-set Region Proposal Network (RPN) is obtained by training an off-the-shelf RPN using the LVIS dataset [25].

Incremental learning model. Our proposed incremental learning framework is evaluated against two existing incremental object detection methods: RILOD and DE-ViT [26]. RILOD collects images through online searching and introduces an automatic image labeling method, which faces scalability challenges and lacks robustness in annotation quality. DE-ViT is a state-of-the-art few-shot object detection model. The 20 novel classes are randomly divided into four sessions, with 5 classes per session. We test four different training settings for novel classes, ranging from 1-shot to 30-shot

per class. In the class-incremental learning process, both real images and generated images are used to assess the generative model’s effectiveness.

Hardware Setup. The object detection models are deployed on a Jetson AGX Orin, while all other models are deployed on a workstation equipped with an RTX 4090D GPU.

B. Experiment Results

Data generation. As shown in Table I, the BK-SDM-small model can generate a single image with accurate annotation in under 1 second. It takes only 20 seconds to generate a 1-session, 5-shot dataset. The visual results of the generated images are shown in Fig.3, where no significant visual difference is observed between the generated and real images. However, as illustrated in Fig.4, the model trained with real images achieves 51% higher accuracy compared to training with generated images. This highlights the key limitation of image generation algorithms. Furthermore, the model accuracy decreases as the number of sessions increases. This drop is due to the limited incremental capacity of the model in our incremental learning approach, which does not involve fine-tuning during the learning process. In contrast, few-shot object detection models are more adaptable to large-scale novel classes, as demonstrated in Table II.

Another advantage of few-shot object detection models is their significantly higher accuracy in detecting novel classes compared to incremental learning models. However, training few-shot object detection models requires either manually annotated images or images generated using Layout-Diffusion, which results in much longer data preparation times. Additionally, while incremental learning models are trained only on the novel class dataset, few-shot object detection requires fine-tuning the entire dataset, including both novel and base classes, which significantly increases model training time. As shown in Fig. 5, DE-ViT incurs four times the data preparation time and thousands of times the model training time. In contrast,

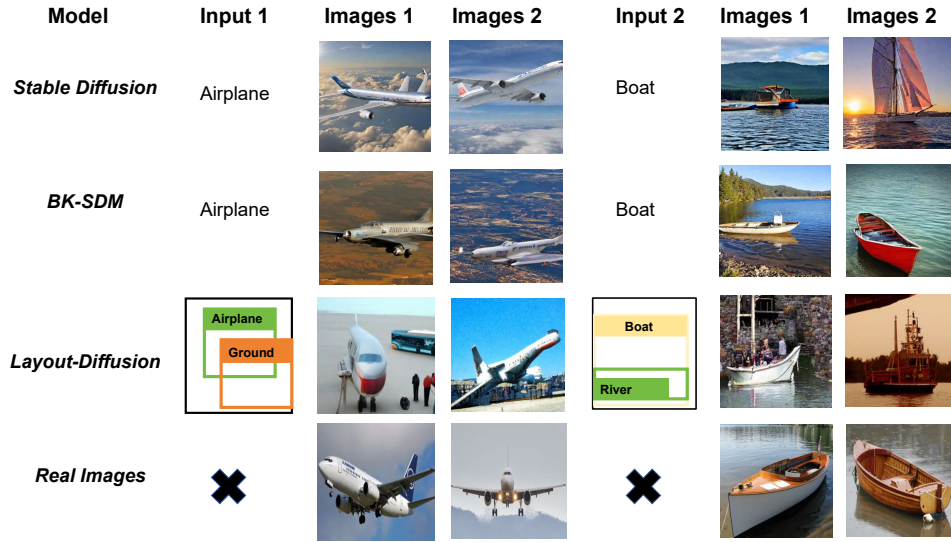


Fig. 3. The visual result of generated images. (a) Stable Diffusion generates images based solely on simple category prompts such as “Airplane” or “Boat”, producing visually appealing scenes but lacking explicit spatial control. (b) BK-SDM also uses textual prompts, with improved structural coherence and object placement compared to (a), though still limited in grounding accuracy. (c) Layout-Diffusion leverages spatial layouts as input (“Airplane” above “Ground”, “Boat” on “River”), guiding the generation process to produce semantically consistent and spatially grounded images. (d) Real Images are provided for reference, showing natural scenes of airplanes and boats, serving as benchmarks for evaluating realism and layout fidelity in the synthetic outputs.

TABLE III
MODEL ACCURACY WITH REAL AND GENERATED DATA AND VARIOUS SESSION AND SHOT NUMBERS

Session	Subset	Real data				Generated data			
		1-shot	5-shot	10-shot	30-shot	1-shot	5-shot	10-shot	30-shot
session 0	base	61.53	61.53	61.53	61.53	61.53	61.53	61.53	61.53
session 1	base	59.40	58.87	55.75	56.17	59.75	59.33	59.01	58.62
	novel	8.20	29.00	37.60	38.20	9.60	21.40	23.00	24.40
	overall	55.14	56.67	54.26	54.61	55.41↑ (+0.27)	55.99	55.93↑ (+1.67)	55.65↑ (+1.04)
session 2	base	58.64	57.72	54.16	54.56	59.14	58.42	57.94	57.76
	novel	3.80	13.50	17.30	17.70	4.40	8.40	9.10	10.40
	overall	54.32	55.31	52.45	52.88	54.78↑ (+0.46)	54.75	54.55↑ (+2.10)	54.56↑ (+1.68)
session 3	base	56.22	55.90	53.10	53.66	58.24	57.41	56.95	56.82
	novel	2.20	7.13	8.93	10.40	2.80	4.67	4.93	5.33
	overall	52.03	53.27	50.86	51.79	53.98↑ (+1.95)	53.65↑ (+0.38)	53.33↑ (+2.47)	53.31↑ (+1.52)
session 4	base	55.38	54.94	52.22	52.96	57.61	56.92	56.51	56.37
	novel	1.45	4.85	6.35	7.30	1.80	3.30	3.50	3.75
	overall	51.21	52.29	49.95	50.92	53.30↑ (+2.09)	53.13↑ (+0.84)	52.81↑ (+2.86)	52.75↑ (+1.83)

our proposed two-stage incremental framework achieves a 10x faster model training speed compared to existing incremental approaches like FLOD.

Incremental learning. Comprehensive experiments were conducted to further investigate the characteristics of the proposed incremental learning model. The experimental results are summarized in Table III. As previously mentioned, real data achieves higher accuracy. Additionally, both base and novel class accuracies decrease with an increase in the number of sessions. However, the accuracy of novel classes improves as the number of image shots increases. Specifically, the 5-shot setting yields up to a 55% accuracy gain compared to the 1-shot setting. Interestingly, further increases in the number of image samples do not necessarily lead to higher

accuracy. As shown in Fig. 6, the accuracy only improves by 6% when increasing from 5-shot to 10-shot and by 5% when increasing from 10-shot to 30-shot. Therefore, the 5-10 shot range represents the most efficient setting for incremental object detection.

Moreover, the relationship between training time and session number in the proposed incremental learning framework is approximately linear, as shown in Fig. 7. For example, 4 sessions (20 novel classes) can be integrated into the existing object detection model within half a minute. This demonstrates the scalability of the incremental learning approach.

V. CONCLUSION

This work presents the first end-to-end real-time open-set object detection framework for MAR applications. By intro-

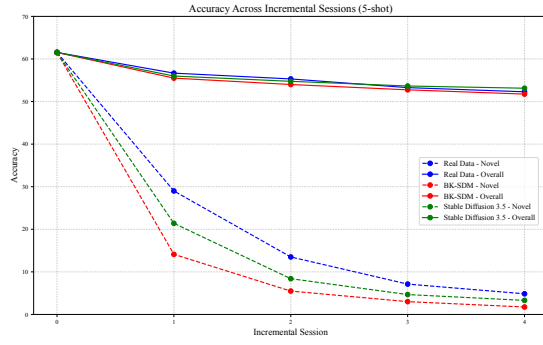


Fig. 4. Impact of incremental session number on model accuracy.

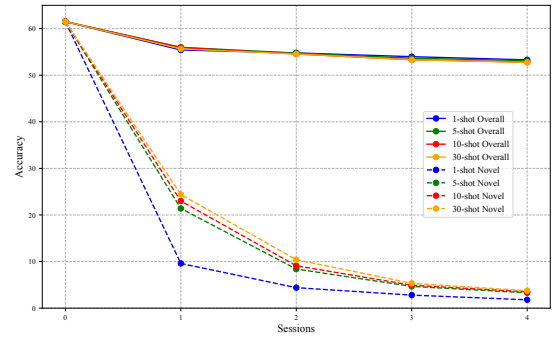


Fig. 6. Impact of shot number on model accuracy.

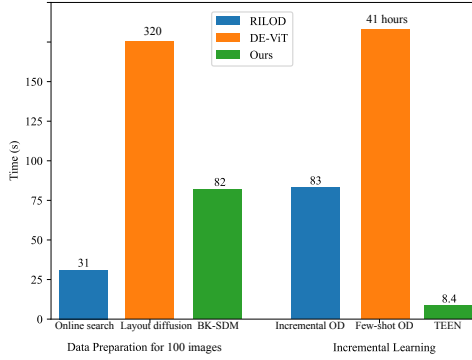


Fig. 5. Time for data preparation and incremental learning.

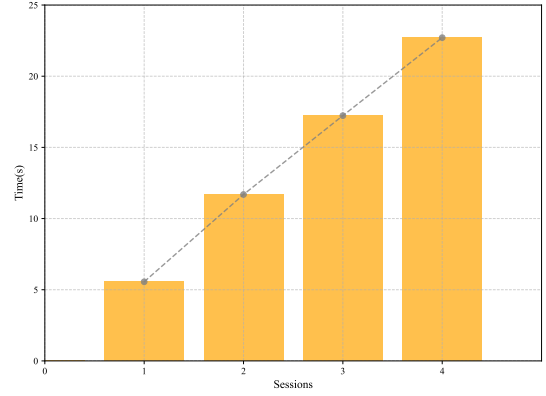


Fig. 7. Impact of session number on model accuracy.

ducing a two-stage object detection model comprising a frozen open-set RPN and a hot classification module, we enable fast and efficient incremental learning. The integration of image generative models, such as Stable Diffusion, facilitates the generation of both images and annotations, further enhancing the framework’s adaptability. Our extensive experimental results demonstrate that the proposed framework significantly accelerates incremental object detection, achieving a 10x faster processing speed, completing the task in just 2 minutes. Additionally, the comprehensive experiments provide valuable insights into the behavior of incremental object detection, offering a deeper understanding of its potential for scalable deployment in MAR applications.

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